Date:

Scaled Dot-Product Attention

1) Explicit Q, K & V :-

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \quad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \quad V = \begin{bmatrix} 1 & 2 \\ 3 & 0 \\ 0 & 4 \end{bmatrix}$$

2) Compose QK^T :-

$$K^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$S_{11} = 1 \quad S_{12} = 0 \quad S_{13} = 1$

$S_{21} = 0 \quad S_{22} = 1 \quad S_{23} = 1$

$S_{31} = 1 \quad S_{32} = 1 \quad S_{33} = 2$

$$QK^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

3) Scaling scores by $\sqrt{d_k}$:-

$d_k = 2$; $S = QK^T/\sqrt{2}$

$$S = \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 0.7071 & 0.7071 \\ 0.7071 & 0.7071 & 1.4142 \end{bmatrix}$$

4) Applying row-wise softmax :-

for each row i & key column j, $A_{ij} = \exp(S_{ij}) / \Sigma_k \exp S(ik)$

row 1 :- $D_1 = 2.0281 + 1 + 2.0281 = 5.0562$

$A_1 = 2.0281/D_1, 1/D_1, 2.0281/D_1 = [0.4011, 0.1978, 0.4011]$

row 2 :- $D_2 = 5.0562$

$A_2 = [0.1978, 0.4011, 0.4011]$

row 3 :- $D_3 = 8.1695$

$A_3 = [0.2483, 0.2483, 0.5035]$

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$$A = \begin{bmatrix} 0.4011 & 0.1978 & 0.4011 \\ 0.1978 & 0.4011 & 0.4011 \\ 0.2483 & 0.2483 & 0.5035 \end{bmatrix}$$

5.) product attention weights by V

$O_1 = [0.9944, 2.4067]$

$O_2 = [1.4011, 2.00]$

$O_3 = [0.9930, 2.5105]$

$$O = \begin{bmatrix} 0.9944 & 2.4067 \\ 1.4011 & 2.0000 \\ 0.9930 & 2.5105 \end{bmatrix}$$

6.) Applying casual mask before softmax

$$M = \begin{bmatrix} 0 & -\infty & -\infty \\ 0 & 0 & -\infty \\ 0 & 0 & 0 \end{bmatrix} \qquad S+M = \begin{bmatrix} 0.7071 & -\infty & -\infty \\ 0 & 0.7071 & -\infty \\ 0.7071 & 0.7071 & 1.4142 \end{bmatrix}$$

$\exp(-\infty) = 0$; forbidden positions = zero weight

row 1 = $[\exp(0.7071)/\exp(.71, 0, 0) = (1, 0, 0)$

row 2 = $[0.3302, 0.6698, 0]$

row 3 = $[0.2483, 0.2483, 0.2483]$

$$A\text{-Masked} = \begin{bmatrix} 1.000 & 0 & 0 \\ 0.3302 & 0.6698 & 0 \\ 0.2483 & 0.2483 & 0.5035 \end{bmatrix}$$

O_1 (masked) = $[1.000, 2.000]$

O_2 (masked) = $[2.3395, 0.6605]$

O_3 (masked) = $[0.9930, 2.5105]$.

$$O\text{-masked} = \begin{bmatrix} 1.000 & 2.000 \\ 2.3395 & 0.6605 \\ 0.9930 & 2.5105 \end{bmatrix}$$

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7) Without a mask, every token may attend to all 3 tokens, including those appearing later in sequence
With a causal mask: token 1 sees only token 1. Token 2 sees token 1 and 2. Token 3 sees tokens 1, 2, 3.
Future key attention is zero, so a GPT style model cannot access tokens that occur after the current position while predicting next token
During teacher-forced training, casual attention can calculate all position outputs in parallel while respecting this restriction. During autoregressive generation, newly generated tokens are still produced one after another.